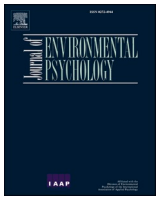
Journal of Environmental Psychology 105 (2025) 102662



Contents lists available at ScienceDirect

Journal of Environmental Psychology

journal homepage: www.sciencedirect.com/journal/journal-of-environmental-psychology





Psychophysiological effects of walking in forests and urban built environments with disparate road traffic noise exposure: A randomized controlled trial

Julia Schaupp [a,c,*], Beat Schäffer [b], Jean-Marc Wunderli [b], Karin Hediger [d,e,f], Silvia Tobias [a], Jakob Heimer [g], Nicole Bauer [a]

[a] Swiss Federal Institute for Forest, Snow and Landscape Research (WSL), Birmensdorf, Switzerland
[b] Empa, Swiss Federal Laboratories for Materials Science and Technology, Dübendorf, Switzerland
[c] University of Basel, Faculty of Psychology, Department of Clinical Psychology and Psychotherapy, Basel, Switzerland
[d] University of Lucerne, Faculty of Behavioural Sciences and Psychology, Department of Child and Adolescent Psychology, Luzern, Switzerland
[e] Swiss Tropical and Public Health Institute, Department of Epidemiology and Public Health, Allschwil, Switzerland
[f] Open University, Faculty of Psychology, Heerlen, the Netherlands
[g] Swiss Federal Institute of Technology Zurich, Department of Mathematics, Seminar for Statistics, Zurich, Switzerland

ARTICLE INFO

Keywords:
Road traffic noise
Greenspace
Restoration
Stress
Attention
Affective responses

ABSTRACT

People in urbanized societies commonly visit greenspaces to alleviate stress and promote restoration and well-being. However, access to greenspaces is often limited and road traffic noise remains a major environmental stressor in urban areas. While various studies have compared the benefits of exposure to greenspaces and urban built environments, the role of traffic noise in shaping these effects remains understudied. This randomized, controlled field study compared the psychophysiological benefits of exposure to forests and urban built environments with different levels of road traffic noise in 354 healthy adults during 30-min walks in Zürich, Switzerland. A range of physiological and psychological outcomes were measured. Linear mixed-effects models showed that walking in forests, in environments with lower road traffic noise and longer relative quiet time without road traffic noise was associated with increased feelings of restoration, subjective stress recovery, and perceived restorativeness, compared to walking in urban built environments with higher road traffic noise as well as such with less relative quiet time. These associations were not observed for salivary cortisol and for attention, though cortisol levels decreased in all conditions, likely due to walking itself. Findings from quantitative and qualitative data on soundscape-induced positive and negative affect showed discrepancies, indicating a need for developing and using affective measures that include low activation affect. Our results suggest that even when noise exposure is relatively low, it can still impede subjective restoration and subjective stress recovery. This highlights the need for noise mapping and mitigation strategies to reduce noise exposure, specifically in public greenspaces. The results further underscore the importance of noise-free periods, without road traffic noise. When planning, designing, and/or preserving urban environments to support the well-being of inhabitants, road traffic noise as an impediment to subjective restoration and subjective stress recovery should be addressed.

1. Introduction

Stress is a fundamental, pervasive, and rather widespread component of everyday life. Chronic stress negatively impacts both psychological and physiological health. The 2022 Swiss Health Survey reveals that the proportion of individuals reporting frequent or constant workplace stress rose from 18 % in 2012 to 23 % in 2022 (Swiss Federal Statistical Office, 2024). Chronic stress increases the risk of gastrointestinal disorders (Black et al., 2022), elevated blood pressure (Landsbergis et al., 2013), diabetes (Hackett & Steptoe, 2017), dementia (Franks et al., 2021), and cardiovascular disease (Osborne et al., 2020). Stress is thus a public health issue, and there is a need for research on stress recovery, stress prevention, and ways of improving psychophysiological well-being.

* Corresponding author. Swiss Federal Institute for Forest, Snow and Landscape Research (WSL), Zuercherstrasse 111, Birmensdorf, 8903, Switzerland.
E-mail address: julia.schaupp@wsl.ch (J. Schaupp).

https://doi.org/10.1016/j.jenvp.2025.102662
Received 27 December 2024; Received in revised form 16 June 2025; Accepted 16 June 2025
Available online 18 June 2025
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Urban greenspace has a long history as a means of promoting public health (Hartig et al., 2011), and the various ways in which it serves as a health resource receive substantial attention (European Environment Agency, 2020b; Markevych et al., 2017). Literature suggests that spending time in greenspaces can alleviate everyday stress and promote restoration and well-being. This has been demonstrated through subjective assessments (such as the Restorative Outcomes Scale; see Korpela et al., 2008; Tyrväinen et al., 2014), shifts in positive and negative affect (McMahan & Estes, 2015), lower levels of the stress biomarker salivary cortisol (e.g. Tsunetsugu et al., 2007), and diminished attentional fatigue (measured by performance on attention-related tests; see Stevenson et al., 2018). However, access to greenspaces is scarce or not available for many living in urban areas. In 2019, the United Nations estimated that 55 % of the world's population live in cities (United NationsDepartmentof Economic and Social Affairs Population Divisio, 2019), with this proportion rapidly growing. Correspondingly, preserving greenspaces as environments conducive for promoting health and well-being is associated with numerous health benefits and opportunities for coping with stress (Beyer et al., 2014; Maas et al., 2006; McCormick, 2017; Roe et al., 2013; Schäffer et al., 2020; Sugiyama et al., 2008; Twohig-Bennett & Jones, 2018; Van Renterghem, 2019; Vienneau et al., 2017, 2017, 2017; Wood et al., 2017).

Numerous studies have examined the beneficial effects of exposure to greenspaces in comparison to urban built environments (see e.g. Bratman et al., 2015; Rozario et al., 2024; Sudimac et al., 2024 for recent field studies). A meta-analysis by Bowler et al. (2010) provided mixed evidence for emotional benefits of exposure to greenspaces compared to built environments indicating a stronger decrease in anger and sadness in natural environments compared to synthetic environments but no difference for decreased anxiety, increased energy, and tranquility. Mostly positive effects of exposure to greenspaces vs. urban built environments on mood were found in a systematic review by Kondo and colleagues (2018). Less evidence was found on benefits for attention, blood pressure, and cortisol (Bowler et al., 2010; Kondo et al., 2018). Bowler and colleagues (2010) highlight the lack of high-quality studies. They discuss the challenge of selecting appropriate environments for comparison when investigating the benefits of natural environments and suggest that differences in benefits between natural and alternative environments may be influenced by additional stressors, such as traffic noise, in the comparison environments. Thus, the benefits of exposure to greenspaces found in earlier studies may be at least partly due to potentially disparate traffic noise exposure in the comparison settings. Few studies have considered noise exposure in the comparison environments (Gidlow et al., 2016; Tyrväinen et al., 2014), but in these studies noise was not sufficiently controlled to avoid confounding. E.g., Gidlow et al. (2016) compared the benefits of walking in natural environments with and without water with a pleasant urban residential environment, aiming at comparable noise exposure in the environments. However, no continuous noise measurements were carried out during the walks. Instead, noise was estimated with two 15-min recordings at representative points in each environment (urban residential 50.6 $\pm$ 4.3 dBA, natural without water 47.5 $\pm$ 2.9 dBA, natural with water 45.6 $\pm$ 1.5 dBA). Given the high temporal variability of traffic noise and its rapid fluctuations (Karipidis et al., 2014), continuous noise measurements would yield substantially more accurate and representative exposure data than isolated short-term recordings.

Indeed, urban areas are typically characterized by high road traffic noise levels (European Environment Agency, 2020a; WHO Regional Office for Europe, 2011). Compared to other environmental risk factors, traffic noise accounts for 24 % of environment-related health loss in the Netherlands (de Hollander et al., 1999) and has one of the largest public health impacts in terms of environmental burden of disease in Western Europe (Hänninen et al., 2014). Negative health impacts of road traffic noise, ranging from e.g. depression (Stobbe et al., 2022) to hypertension (van Kempen & Babisch, 2012) and blood pressure (Lee et al., 2019) have been reported. Evidence suggests associations between noise exposure, noise annoyance and stress. Reduced long-term road traffic noise exposure has been linked to reduced annoyance and enhanced wellbeing (Ohrström, 2004). Further, skin conductance was shown to increase with road traffic noise increasing from 50 to 70 dBA (Ellermeier et al., 2019). Finally, performing a cognitive task in a louder environment (90 dBA) raised salivary cortisol levels more compared to a quieter setting (45 dBA) (Tafalla & Evans, 1997). However, most previous related findings rely on laboratory research; field experiments have revealed weaker effects and frequently relied on small sample sizes (Barbaresco et al., 2019). Unlike the aforementioned studies, the present research does not focus on the detrimental health effects of noise, but rather on how noise interferes with restoration—specifically, how it may hinder or reduce restorative experiences. A related laboratory study by Uebel et al. (2021) demonstrated that perceived restorativeness declined as perceived road traffic noise levels increased.

Given the limitations in existing research, there is a clear need for high-quality field studies investigating the acute effects of objectively measured traffic noise on stress and restoration. The present study addresses this gap by examining the psychophysiological benefits of exposure to greenspaces versus urban built environments with minimal road traffic noise confounding. In addition, it has been suggested that focusing solely on averaged noise metrics ignores other potentially important aspects of a noise situation, such as its macro-temporal pattern (Guski, 1985; Schäffer et al., 2022). Indeed, the duration of quiet periods in between events of road traffic noise have been associated with lower noise annoyance (Brink et al., 2019; Schäffer et al., 2022). Consequently, the present study used a randomized, controlled field experiment in ecologically valid settings to compare the acute benefits of walking in greenspaces and urban built environments with different road traffic noise exposure, measured by averaged A-weighted sound pressure levels as well as relative quiet time. Employing a factorial design, it examined the individual contributions of the walking environment and road traffic noise. Unlike previous studies that deliberately exposed participants to a stressor at the outset (Berman et al., 2008; Hartig et al., 2003; Hedblom et al., 2019; Ulrich et al., 1991), this research focused on healthy adults that were not deliberately stressed before the intervention, exploring potential improvements for individuals with levels of stress close to those present in everyday life. This is relevant because greenspace exposure may offer benefits beyond recovery from high stress or restoring depleted directed attention. Processes of restorative potential without a preceding stress intervention have been called "instorative" (Hartig, 2007; Hartig et al., 1996; Korpela & Ratcliffe, 2021). Accordingly, the present study examines the instorative benefits of walking in different environments. For the exact research questions and hypotheses see Supplement A.

2. Material and methods

2.1. Participants

Fig. 1 provides information on the participant flow, including exclusions with reasons. A total of 354 healthy adults in 97 groups were included in the final analysis. Individuals were eligible if they: (a) were at least 18 years old, (b) could walk at a moderate pace for 30 min, (c) had no diagnosed hearing impairments, (d) did not use cortisone for medical reasons, and (e) had a BMI below 35. To ensure a broad age range, a random sample of 7000 residents was drawn from the Zurich registration office, stratified by four age groups: 18–35, 35–50, 50–65, and over 65. Additional participants were recruited via universities, local newspapers, community web portals, and public flyers. Participants received 50 CHF (approx. 50 EUR) compensation. A description of the conducted power analysis is provided in Supplement B.

2.2. Procedure

Participants took part in a guided group walk with up to five

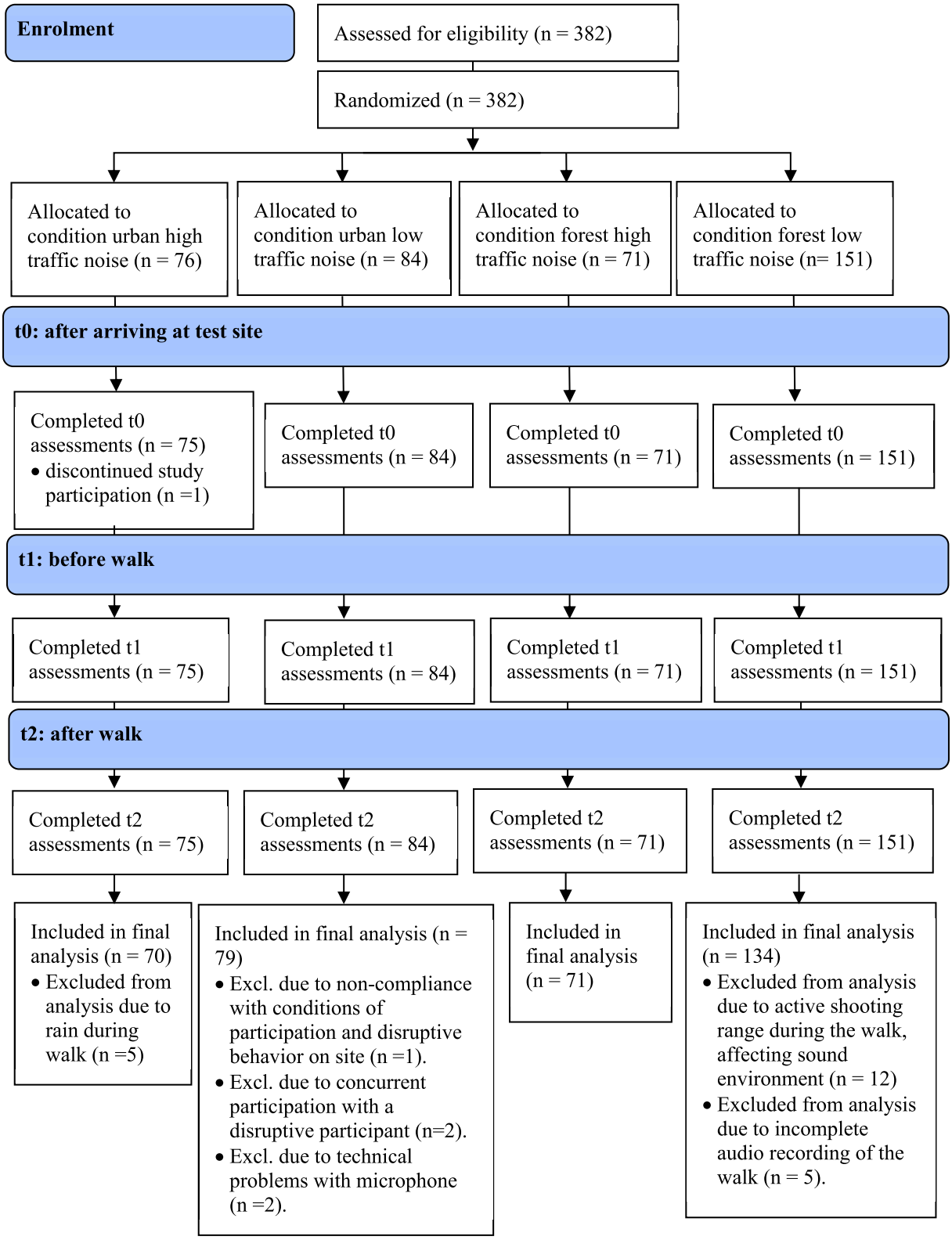


Fig. 1. Participant flow diagram with number of included and excluded participants.

participants. The participants were randomly assigned to one of four conditions (urban built environment or forest, with high or low road traffic noise). Assessments took place before (t1) and after (t2) the walk. The study was preregistered (trial registration number ISRCTN48943261) and approved by the Ethics Committee of the Swiss Federal Institute of Technology Zürich, Switzerland (Ref No. EK 2021-N-211 of January 27, 2022). See Fig. 2 for a flow chart on the study procedure.

Participants were instructed to avoid caffeine and strenuous physical activity for 2 h prior to their walking appointment, walking activity on the day of the appointment, and drug use for 24 h prior to their session. Upon arrival at the test site, participants received a brief introduction to

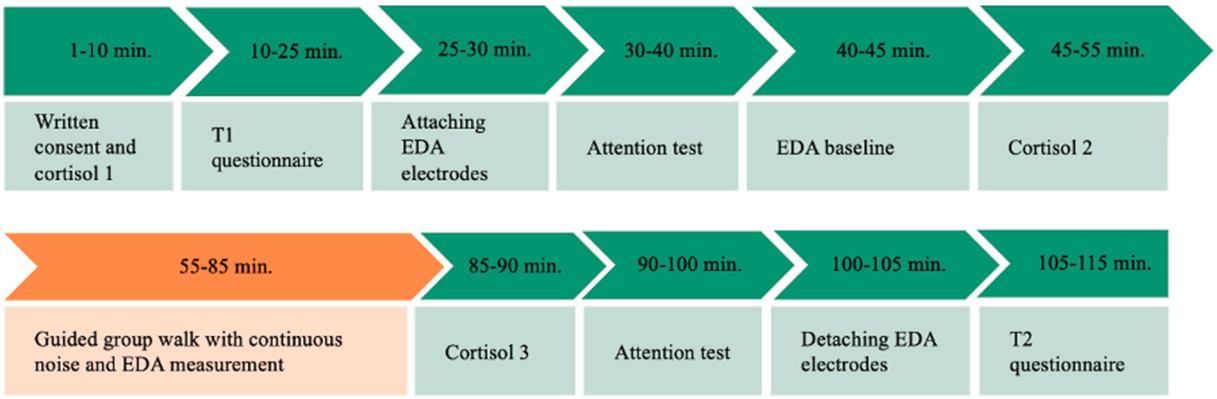


Fig. 2. Flow chart of study procedure. Adapted from Schaupp et al. (2024). Data on skin conductance will be reported elsewhere.

the study's research topic, procedures, and privacy policies, and provided written informed consent. Baseline salivary cortisol levels were assessed, followed by questionnaires and an attention test. Before the beginning of the walk, salivary cortisol was assessed with a second probe. Participants were then led for a walk along a predetermined route at a pace of approx. 4.7 km/h. Sound was measured continuously during the walk by the experimenter carrying a calibrated sound level meter (XL2 by NTI Audio, Schaan, Liechtenstein) using a free-field measurement microphone, fulfilling class 1 environmental requirements according to IEC 61672. Participants were instructed not to talk and to maintain at least 6 m distance from the experimenter to prevent their footsteps from affecting noise recordings. After the walk, cortisol was measured again, the attention test was repeated, and participants completed a second questionnaire. See the full questionnaire at https://osf.io/2bfgv/. Data collection occurred in 2022, 2023, and 2024 during weekdays from 1:30 p.m. to 6:00 p.m., when salivary cortisol levels are more stable (Dickerson & Kemeny, 2004; Hansen et al., 2008; Ljubijankić et al., 2008; Von Dawans et al., 2011). A detailed description of the procedure, and methods was published in the study protocol (Schaupp et al., 2024).

2.3. Study sites

Participants walked in one of four environment types in the city of Zürich, Switzerland. For every type, three sites were selected, resulting in 12 test sites. Fig. 3 depicts example pictures of each condition.

A description of the test site selection process is given in Supplement C. For the exact locations of the starting points of the walking routes in the test sites and the GPS files of the walking routes see https://osf.io/2bfgv/.

The urban high traffic noise walks took place along sidewalks next to a street with intense road traffic, while the low traffic noise walks went along sidewalks beside less busy streets in residential areas. In forests, high traffic noise walks went along sidewalks and paths next to a road crossing the forest, and low traffic noise walks followed footpaths with no or minimal road traffic noise. These sites reflect common urban built and urban forest environments in Zürich and were chosen due to the practical relevancy of these environments for residents as possible places to take a walk. However, they are not necessarily representative of such environments in general. Table 1 shows the acoustic characteristics in the different conditions as assessed during the walks.

Table 1 displays that average exposure duration was similar between conditions (28–31 min) with a slightly shorter average exposure duration in the forest with low traffic noise condition. Sound exposure level (L_{AE}) for times during which traffic noise was dominant was highest in the urban built condition with high road traffic noise, followed by the forest condition with high road traffic noise, and was lowest in the forest condition with low road traffic noise. Generally, road traffic noise was higher in urban built, compared to forest conditions. The results of the noise measurements during the walks are consistent with the subjective reports of the participants regarding the extent to which traffic noise was heard, unpleasantness of the sound environment, and annoyance due to road traffic noise, which all depicted the highest scores in the urban condition with high traffic noise and the lowest score in the forest condition with low traffic noise. Information on the acoustic and vegetation details across all test sites, as well as details on the forest composition across the forest test sites are provided in supporting information C and the study protocol.

2.4. Measures

The primary outcomes were a stress biomarker and perceived restoration. Where available, validated German versions of the questionnaires were used (for full questionnaire visit https://osf.io/2bfgv/). For a more detailed description of the measures, including reliability, see supporting information D.

2.4.1. Primary outcomes

Perceived restoration. Perceived restoration was measured with an adapted version of the ROS (Korpela et al., 2008).

Stress biomarker. Salivary cortisol was used as a biomarker of stress. The sampling and measurement procedure is detailed in Schaupp et al. (2024). There is a time lag in the salivary cortisol response of 5–20 min after the onset of a stressor, and cortisol levels reach a maximum 10–30 min after stress cessation (Kirschbaum & Hellhammer, 2007). Therefore, we used the mean cortisol value from the first and second cortisol probes as the pre-walk stress measure and the third value as the post-walk stress measure.

2.4.2. Secondary outcomes

Positive and negative affect. Positive and negative affect was assessed with the Positive and Negative Affect Schedule (PANAS) (Watson et al., 1988), which measures positive (e.g., "active" and "enthusiastic") and negative (e.g., "nervous" and "distressed") affect on two separate scores with 10 items each.

Perceived restorative qualities of environment. The restorative qualities of the environment were assessed with an adaptation of the Perceived Restorativeness Scale-11 (PRS-11) (Pasini et al., 2014; cf. Hartig et al., 1997).

Stress recovery. Participants indicated to what extent the environment they walked through contributed to their being able to recover from stress by using a 7-point Likert-like scale ranging from 1 (*not at all*) to 7 (*completely*).

Attention. Attention was measured using the Necker Cube Pattern Control Test (NCPCT) (Cimprich, 1993; De Young, 2022). The test constitutes a three-dimensional wireframe drawing of a cube without



Fig. 3. Examples of the test site conditions. (a) Urban built high traffic noise, (b) urban built low traffic noise, (c) forest high traffic noise, (d) forest low traffic noise.

depth information that can be viewed from two perspectives, with spontaneous reversals between these perspectives occurring with sustained viewing (De Young, 2022). The average number of reversals between two 1 min runs before and then again after the walk was used as an outcome value (Hartig et al., 2003). A smaller number of perspective switches is interpreted as higher directed attention.

Open-ended question. For more insight in participants' affect, an open-ended question related to the feelings evoked by the soundscape was included (*How did the soundscape during the walk make you feel?)*.

2.4.3. Objective road traffic noise measures

Audio files of sound continuously measured during the walks were listened to and categorized per 1-s period into road traffic noise being the dominant (or non-dominant) sound source. We then calculated the corrected sound exposure level (L_{AE}) for the cumulative time periods in which traffic noise was the dominant sound source, including times when road traffic noise was not dominant with a corrected preset noise exposure (L_{Aeq}) of 30 dBA.

The cumulative duration in seconds in which traffic noise was dominant was calculated as

$$L_{AE} = L_{Aeq} + log_{10}(T)$$

Relative quiet time (RQT) was defined as the ratio of the total amount of time during which traffic noise was not dominant ($T_{Traffic\text{-}NoiseNotDom}$) to the total duration of the walk (T_{Walk}), and was calculated as

Table 1
Acoustic characteristics in the different conditions during the walks.

	UBHT	UBLT	FHT	FLT
	n = 70 M (SD)	n = 79 M (SD)	n = 71 M (SD)	n = 134 M (SD)
Duration of walks (min.)	31.40 (1.75)	30.40 (1.43)	30.32 (1.82)	28.78 (1.42)
L_{AE} (dBA) on walking route when traffic noise was dominant[a]	98.80 (1.54)	89.87 (5.31)	93.26 (1.97)	76.44 (5.25)
L_{Aeq} (dBA) on walking route when traffic noise was dom.[a]	66.05 (1.62)	57.27 (5.34)	60.67 (2.10)	44.07 (5.25)
RQT (%)[b]	10.50 (8.72)	39.07 (16.00)	15.48 (15.28)	71.87 (23.26)
Extent to which traffic noise was heard[c]	4.21 (0.70)	3.37 (0.82)	4.09 (0.65)	2.33 (0.95)
Unpleasantness of sound environment[d]	7.21 (2.06)	4.68 (2.24)	6.36 (2.35)	2.95 (2.58)
Road traffic noise annoyance[e]	9.00 (2.05)	6.28 (3.18)	8.87 (2.30)	2.11 (1.92)

[a] With periods when road traffic noise was not dominant defined by a preset noise exposure (LAeq) of 30 dBA, based on classifying each 1-s segment of noise recordings into road traffic noise being the dominant (or non-dominant) sound source.
[b] Relative quiet time (RQT): Proportion of time during which traffic noise was not dominant (dom.), ranging from 0 to 100 %.
[c] On a Likert scale ranging from 1 *(not at all)* to 5 *(completely dominating)*.
[d] On a Likert scale ranging from 1 *(very pleasant)* to 10 *(very unpleasant)*.
[e] On a Likert scale ranging from 1 *(not at all)* to 11 *(extremely)*.
UBHT = urban built high traffic noise, UBLT = urban built low traffic noise, FHT = forest high traffic noise, FLT = forest low traffic noise.

$$\mathrm{RQT}\ (\%) = 100 * \frac{T_{TrafficNoiseNotDom}}{T_{Walk}}$$

2.4.4. Additional subjective acoustic measures

Acoustic impression of the soundscape. The impression of the acoustic environment was assessed following the standard ISO/TS 12913-2 (International Organization for Standardization, 2019). The extent to which participants heard traffic noise, sounds from humans, natural sounds, and other sounds was rated on a 5-item Likert-like scale ranging from 1 (*not at all*) to 5 (*dominates completely*). In addition, participants rated the quality of the overall sound environment during the walk on a 10-point Likert-like scale ranging from *very pleasant* to *very unpleasant*.

Noise annoyance. Noise annoyance was measured using the numeric 11-point scale developed by the International Commission on Biological Effects of Noise (Fields et al., 2001).

2.5. Data analyses

2.5.1. Quantitative analyses

The outcomes were analyzed with linear mixed-effects models. For outcomes measured only post-walk, models comparing the post-walk score across conditions were calculated. For outcomes assessed before and after the walk, models comparing change scores were conducted. Positive cortisol delta scores (before – after) reflect a decrease in stress. Positive NCPCT delta values (before – after) display increased attention. Positive delta positive affect scores (before – after) indicate a decrease in positive affect, while positive delta negative affect scores represent decreased negative affect.

No variable showed missing values $\geq$ 5 %. Little's missing completely at random (MCAR) test was nonsignificant (p = 0.087), indicating that MCAR can be inferred. Missing data were dealt with using maximum likelihood estimates.

In all models, fixed effects for the walking environment and noise as well as an interaction term between environment and the noise predictor were specified. All models included age as a covariate, as previous research suggests older respondents being more annoyed by noise (Brink et al., 2019) and, e.g., greater effects of exposure to natural environments on positive affect for older participants (Yao et al., 2021). In addition, gender was included as a covariate in all models. The group participants had walked in, was added as a random intercept to account for the hierarchical structure of the data. For negative affect and attention, within-group dependency of observations was low, so models did not converge. For these outcome variables, linear models without random effects were computed. For each outcome variable, a first model with walking environment and noise as categorial predictors (forest vs. urban built; high vs. low road traffic noise) was calculated. This model was formulated as:

$$lmer\big((outcome) \sim environment_{dummy} \times noise_{dummy} + age + gender + (1\,|group), \mathrm{data} = \mathrm{data}\big).$$

Participants who identified as 'other' gender (n = 3) were excluded from models with noise as a dummy-coded predictor due to their limited number and absence in some conditions. A second model incorporated noise as a continuous predictor using sound exposure level (L_{AE}), formulated as follows:

$$lmer\big((outomce) \sim environment_{dummy} \times L_{AE} + \mathrm{age} + \mathrm{gender} + (1\,|\mathrm{group}), \mathrm{data} = \mathrm{data}\big)$$

In addition, a third model included RQT as another potentially important noise parameter, instead of the L_{AE} formulated as:

$$lmer\big((outomce) \sim environment_{dummy} \times RQT + age + gender + (1\,|\,group), \mathrm{data} = \mathrm{data}\big)$$

For more details on the model formulations see Supplement E. We accounted for multiple testing by adjusting the p-values for the number of post hoc calculations computed separately within every model by using the Holm method (Holm, 1979).

Initially, we aimed to include both continuous noise predictors (L_{AE} and RQT) in a single mixed-effects model. However, due to high correlation (variance inflation factor VIF >5), we included them in separate models. All analyses were performed with a significance level set at 0.05 using R version 4.3.3 (R Core Team, 2018) and R package lme4 (Bates et al., 2015). For models with a categorical noise predictor, we analyzed the effect of environment independently of noise, as well as the effect of noise independently of environment, using post hoc custom contrasts with the R package emmeans (Lenth et al., 2024). For the models with continuous noise predictors, we calculated post hoc marginal effects comparing forest and urban built environment at the mean value of noise and comparing environments with higher and lower traffic noise/RQT across walking environment types in light of nonsignificant interactions. The association of L_{AE}/RQT and the outcomes was analyzed separately for the environment conditions (urban built/forest), ignoring the interaction and calculating post hoc marginal effects of L_{AE}/RQT while the environment predictor was set to the specific level (0 = urban, 1 = forest).

2.5.2. Qualitative data analyses

For deeper insight into participant experiences, responses to the open-ended question on soundscape-induced feelings were

systematically analyzed as follows: First, emotions were categorized as "positive," "negative," "neutral," "mixed/unclear valence," "none," or "other than feelings stated." We used an inductive category formation approach (Mayring, 2015) to identify categories of most frequently occurring feelings caused by the soundscape, separately for the conditions participants had walked in. Similar terms (e.g., relaxed, relaxation) were grouped as the same affect. Thereafter, the five most frequently mentioned emotions were deductively coded on the two dimensions valence and arousal based on the circumplex model of affect (Russell, 1980). To compare the freely reported affects with the responses on the closed affect question in the questionnaire, the PANAS items were likewise arranged according to the circumplex model of affect. Three authors independently mapped the PANAS affects onto the dimensions of valence and arousal. When deviations occurred, the classification with the highest level of agreement was selected.

3. Results

3.1. Results from quantitative data

Table 2 summarizes demographic characteristics of the included participants.

Table 3 displays the means and standard deviations for the outcome variables by condition across measurement time points for the models with noise as categorial predictor. Descriptively, restoration, perceived restorativeness, and subjective stress recovery were higher after walking in forests than in urban built environments and higher in environments with low traffic noise, compared to such with high traffic noise. Cortisol and negative affect descriptively decreased in all conditions. Descriptively, attention increased in the condition of forest with low traffic noise and the condition of urban built with high traffic noise. Positive affect descriptively increased in all conditions except urban built with high traffic noise.

3.1.1. Results primary outcomes

Tables 4 and 5 and Fig. 4 show the results of the mixed-effect models with road traffic noise included as categorial predictor (high vs. low exposure) for the primary outcomes.

Table 4 shows a significant marginal effect of environment and noise on the restoration. Restoration was higher after walking in forests, compared to walking in urban built environments (b = 0.982, SE = 0.173, p < .001, CI(0.637, 1.327)), and higher after walking in environments with low road traffic noise, compared to walking in environments with high traffic noise (b = −0.757, SE = 0.172, p < .001, CI (−1.099, −0.416)). No significant interaction between environment and noise was observed.

Table 5 indicates that for salivary cortisol, there was no significant marginal effect of environment or noise. Additionally, the interaction between environment and noise was not significant.

As displayed in Tables 4 and 5, age was a significant covariate for both cortisol (b = −0.012, SE = 0.004, p = 003, CI(−0.020, −0.005)), and restoration (b = −0.013, SE = 0.004, p = .001, CI(−0.021, −0.005)), while gender was not. Restoration was lower in older participants and cortisol decreased less in older participants, compared to younger ones.

The results of the mixed models with road traffic noise as a continuous variable (L_{AE}) (Tables 1 and 2 as well as Fig. 1 in Supplement F) as well as the results of the mixed models with RQT (Tables 8 and 9 and Fig. 3 in Supplement F) are largely consistent with those reported for the models with dummy-coded noise regarding the direction of the effects for the primary outcomes (see Table 11 on the results of all three models). The only inconsistency was that, unlike the other models, the model with L_{AE} indicated that compared to walking in urban built environments with low road traffic noise, walking in urban built environments with high road traffic noise did not lead to significantly lower restoration (b = −0.035, SE = 0.022, p = 0.111, CI(−0.077, 0.008)). The effect of noise averaged across environment conditions was significant (b = −0.044, SE = 0.012, p < .001, CI(−0.067, −0.021)).

3.1.2. Results secondary outcomes

Tables 6–10 and Fig. 5 show the results of the mixed-effect models with road traffic noise included as categorial predictor for the secondary outcomes.

Table 6 indicates a significant marginal effect of both environment and noise on PRS-11. Perceived restorativeness was higher after walking in forests compared to walking in urban built environments (b = 1.207, SE = 0.147, p < .001, CI(0.915, 1.498)), and higher after walking in environments with low road traffic noise compared to walking in environments with high traffic noise (b = −0.795, SE = 0.145, p < .001, CI (−1.084, −0.506)). No significant interaction between environment and noise was found.

Table 7 shows a significant marginal effect of both environment and noise on subjective stress recovery. Stress recovery was higher after walking in forests, compared to walking in urban built environments (b

Table 2
Demographics at baseline.

	UBHT	UBLT	FHT	FLT	Total
	n = 70	n = 79	n = 71	n = 134	n = 354
Age in years, M (SD)	42.13 (18.71)	45.75 (19.11)	41.17 (18.15)	38.13 (18.00)	41.24 (18.57)
Female gender, n (%)	54 (77.1 %)	55 (69.6 %)	47 (66.2 %)	78 (58.2 %)	234 (66.1 %)
Male gender, n (%)	14 ((20.0)	24 (30.4)	23 (32.4)	54 (40.3)	115 (32.5)
Other gender, n (%)	2 (2.9)	0 (0)	0 (0)	1 (0.7)	3 (0.8)
Main occupation, n (%)*					
Employed (full-time)	9 (12.9 %)	15 (19.0 %)	15 (21.1 %)	20 (14.9 %)	59 (16.7 %)
Employed (part-time)	33 (47.1 %)	34 (43.1 %)	24 (33.8 %)	50 (37.3 %)	141 (39.8 %)
In pension	13 (18.6 %)	16 (20.3 %)	14 (19.7 %)	23 (17.2 %)	66 (18.6 %)
Homemaker	1 (1.4 %)	3 (3.8 %)	1 (1.4 %)	6 (4.5 %)	11 (3.1 %)
Unemployed	2 (2.9 %)	2 (2.5 %)	5 (7.0 %)	3 (2.2 %)	12 (3.4 %)
In training/studying	22 (41.4 %)	25 (31.6 %)	18 (25.4 %)	55 (41.0 %)	120 (33.9 %)
Highest education level n (%)					
Primary/secondary	1 (1.4 %)	3 (3.8 %)	1 (1.4 %)	3 (2.2 %)	8 (2.3 %)
Vocational school	10 (14.3 %)	15 (19.0 %)	9 (12.7 %)	14 (10.4 %)	48 (13.6 %)
High school	20 (28.6 %)	6 (7.6 %)	15 (21.1 %)	43 (32.1 %)	84 (23.7 %)
Higher technical or vocational	10 (14.3 %)	11 (13.9 %)	8 (11.3 %)	7 (5.2 %)	36 (10.2 %)
Univ. of appl. sciences/teacher training	9 (12.9 %)	10 (12.7 %)	9 (12.7 %)	19 (14.2 %)	47 (13.3 %)
Technical college/university	20 (28.6 %)	34 (43.0 %)	27 (38.0 %)	46 (34.3 %)	127 (35.9 %)

Note. UBHT = urban built high traffic noise, UBLT = urban built low traffic noise, FHT = forest high traffic noise, FLT = forest low traffic noise. *Multiple answers possible.

Table 3
Means and standard deviations of outcome variables across measurement time points by condition.

	Before				After			
	UBHT	UBLT	FHT	FLT	UBHT	UBLT	FHT	FLT
	M (SD)	M (SD)	M (SD)	M (SD)	M (SD)	M (SD)	M (SD)	M (SD)
Restoration Outcome Scale					3.38 (1.63)	4.06 (1.28)	4.26 (1.54)	5.23 (1.09)
Salivary cortisol in nmol/L[a]	3.22 (1.81)	2.97 (1.73)	3.41 (1.70)	3.30 (2.00)	2.08 (1.23)	1.96 (1.02)	2.31 (1.41)	2.04 (1.10)
Perceived restorativeness					3.04 (1.04)	3.80 (1.03)	4.22 (1.33)	5.04 (0.96)
Stress recovery in environment					2.76 (1.57)	4.06 (1.51)	4.23 (1.75)	5.74 (1.11)
Positive affect	2.97 (0.74)	2.93 (0.68)	3.02 (0.61)	2.91 (0.65)	2.94 (0.82)	3.03 (0.77)	3.13 (0.77)	3.11 (0.74)
Negative affect	1.60 (0.58)	1.60 (0.55)	1.58 (0.63)	1.60 (0.63)	1.31 (0.44)	1.27 (0.39)	1.28 (0.34)	1.25 (0.35)
Attention (NCPCT)	7.28 (3.66)	8.09 (4.61)	7.86 (4.25)	7.19 (4.63)	6.78 (3.94)	8.21 (4.82)	8.14 (6.03)	6.65 (4.47)

[a] Salivary cortisol scores before the walk are mean scores of the two probes obtained before the walk (first probe taken directly when participants arrived at the test site and second probe taken right before the walk started). UBHT = urban built with high traffic noise, UBLT = urban built with low traffic noise, FHT = forest with high traffic noise, FLT = forest with low traffic noise. Range of possible scores: Restoration Outcome Scale (1–7), perceived restorativeness (1–7), stress recovery in environment (1–7), positive and negative affect (1–5).

Table 4
Effects for restoration outcome scale.

	Estimate	SE[b]	95 % CI[c]	p
Effect of environment averaged across noise conditions[a]	0.982	0.173	0.637, 1.327	<0.001
Effect of noise averaged across environment conditions	−0.757	0.172	−1.099, −0.416	<0.001
Effect of noise in urban built conditions	−0.653	0.256	−1.146, −0.160	0.013
Effect of noise in forest conditions	−0.862	0.229	−1.304, −0.420	<0.001
Age	−0.013	0.004	−0.021, -−0.005	0.001
Gender (male)	0.020	0.154	−0.277, 0.323	0.894
Interaction environment: noise	−0.209	0.343	−0.871, 0.453	0.544

Reference category for environment = urban, reference category for noise = low traffic noise. Reference category for gender = female.
[a] Calculated using custom contrasts.
[b] Standard error.
[c] CI = Confidence Interval.

Table 5
Effects on change in salivary cortisol.

	Estimate	SE[b]	95 % CI[c]	p
Effect of environment averaged across noise conditions[a]	0.061	0.173	−0.284, 0.405	>0.999
Effect of noise averaged across environment conditions	0.001	0.172	−0.341, 0.343	>0.999
Effect of noise in urban built conditions	0.084	0.257	−0.412, 0.579	0.746
Effect of noise in forest conditions	−0.081	0.227	−0.522, 0.358	0.721
Age	−0.012	0.004	−0.020, −0.005	0.003
Gender (male)	0.068	0.161	−0.246, 0.380	0.672
Interaction environment: noise	−0.165	0.343	−0.827, 0.497	0.632

Reference category for environment = urban, reference category for noise = low traffic noise. Reference category for gender = female.
[a] Calculated using custom contrasts.
[b] Standard error.
[c] CI = Confidence Interval.

= 1.531, SE = 0.196, p < .001, CI(1.141, 1.920)), and higher after walking in environments with low road traffic noise, compared to walking in environments with high traffic noise (b = −1.358, SE = 0.194, p < .001, CI(−1.745, −0.972)). No significant interaction between environment and noise was detected.

Table 8 indicates that neither environment nor noise had a significant marginal effect on positive affect. Furthermore, the interaction between environment and noise on positive affect was not significant.

Table 9 shows that neither a significant marginal effect of environment, nor of noise, nor of the interaction between environment and noise on negative affect was observed.

Table 10 indicates that neither environment, nor noise, nor the interaction between environment and noise had a significant marginal effect on attention.

Tables 6–10 further show that age was a significant covariate for negative affect (b = −0.005, SE = 0.001, p < .001, CI(−0.007, −0.002)), stress recovery (b = −0.013, SE = 0.004, p = .002, CI(−0.021, −0.005)), and attention (b = −0.024, SE = 0.009, p = .011, CI(−0.042, −0.005)), but not for positive affect and perceived restorativeness. Negative affect decreased less in older participants. Stress recovery was lower in older participants, and attention increased less in older participants. Gender did not influence any of the secondary outcomes.

The results of the models with road traffic noise as a continuous variable (L_{AE}) (Tables 3–7 and Fig. 2 in Supplement F) as well as the results of the mixed models with RQT (Tables 10–14 and Fig. 4 in Supplement F) are largely consistent with those reported for the models with dummy-coded noise regarding the direction of the effects for the secondary outcomes except for results regarding attention. Table 11 gives an overview of the results of all three models with the different noise predictors.

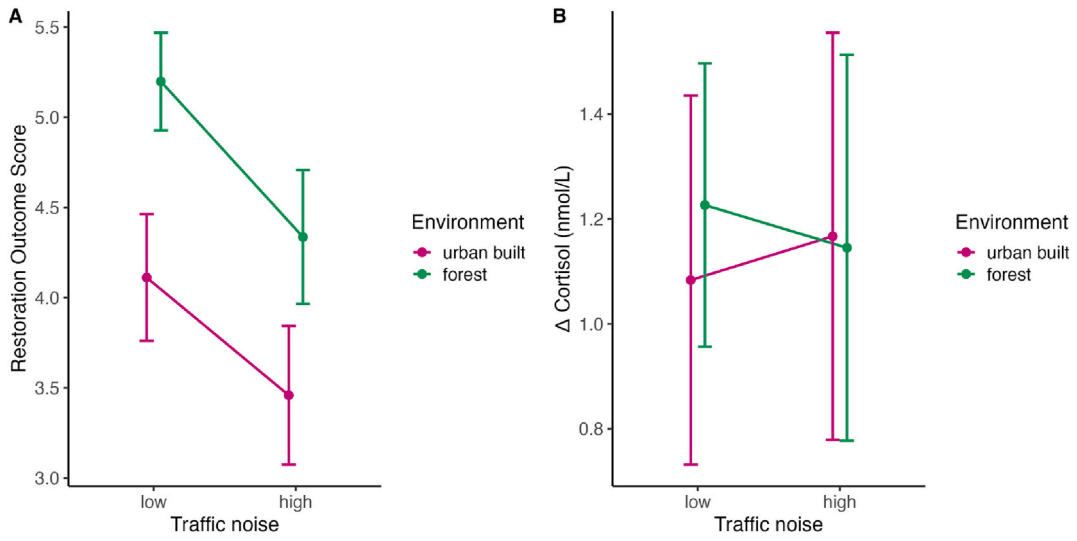


Fig. 4. Estimates for primary outcomes Restoration Outcome Scale and change in cortisol.

3.2. Results from qualitative data

Table 12 shows the five most frequently mentioned affects caused by the soundscape during the walk in the different conditions. Fig. 6 presents the five most frequently mentioned affects arranged according to the circumplex model of affect by Rusell (1980) separately for the different conditions. Fig. 7 displays the PANAS items arranged along the dimensions valence and arousal.

4. Discussion

4.1. Effects of environment and road traffic noise on instoration

Walking in forests conferred greater benefits for restoration, perceived restorativeness, and subjective stress recovery than walking in urban built environments. Our finding of higher restoration following visits to greenspaces aligns with previous studies using the ROS (e.g., Beil & Hanes, 2013; Gidlow et al., 2016; Tyrväinen et al., 2014). Likewise, higher perceived restorativeness following visits to greenspaces, compared to urban built environments, is consistent with recent studies using PRS-11 (e.g., Rozario et al., 2024; Sudimac et al., 2024; Tyrväinen et al., 2014) as well as with results from a meta-analysis (Menardo et al., 2021). Traffic noise was found to be negatively associated with restoration, perceived restorativeness and subjective recovery from stress, supporting Uebel et al. (2021) who reported that higher perceived traffic noise is associated with both perceived restorativeness and restoration. Going beyond the findings of Uebel et al. (2021) on the effects of perceived road traffic noise, our study demonstrated the effects of objectively measured exposure levels as well as the importance of relative quiet time. This aligns with the constrained restoration concept (Hartig et al., 2007; von Lindern et al., 2016), which suggests that certain environmental factors can hinder the restoration process. Road traffic noise thus constrains some beneficial effects of exposure to greenspaces, consistent with previous literature (García-Martín et al., 2025). When the associations of environment and road traffic noise with restoration, perceived restorativeness, and subjective stress recovery are compared, both showed similar strength, though the effect of the environment was slightly more pronounced.

Practically, our results indicate that road traffic noise can impede subjective restoration and subjective stress recovery in both urban forests and urban built environments. When planning, constructing, and preserving environments to support the well-being of inhabitants, the negative effects of road traffic noise on subjective restoration and subjective stress recovery should be addressed. In this respect, noise mitigation strategies are key instruments in reducing citizens' exposure to road traffic noise and thus increase possibilities for restoration. Dynamic noise mapping can provide real-time information to the public via web tools and allows public administrations to take immediate actions against excessive noise exposure by means such as traffic control and management (Asdrubali & D'Alessandro, 2018). For areas of dense traffic, noise mitigation and traffic reduction strategies should be a key focus. Traffic control measures such as flow (Yadav et al., 2022) and speed limits (Brink et al., 2022; Yan et al., 2023) as well as type of vehicles and the use of low-noise road surfaces (Praticò & Anfosso-Lédée, 2012; Teti et al., 2020; Zarei et al., 2022) can be used for effective noise control. Policies might consider action strategies specifically near urban forests, where people go for purposes of restoration, to maintain the positive effect of exposure to such environments. In many countries, the existing Noise Abatement Regulations do not legally apply to recreational areas. As these areas are becoming increasingly important for the recreation of the population due to residential development, noise protection for recreational areas should be incorporated into law in the future. Previous research has proposed a threshold of 45 dB LAeq for daytime transportation noise for greenspaces compatible with restoration (Guinaudeau et al., 2023), as well as a Lday <50 dB(A) for traffic noise exposure in suburban green areas and city parks according to Swedish guidelines (Nilsson & Berglund, 2006). Policies aimed at reducing traffic, such as encouraging the use of bicycle lanes and public transportation, may also be effective. Additionally, a long-term shift to electric vehicles could help mitigate traffic noise. In areas where noise cannot be further reduced, the design of the topography and the planting of shrubs or trees should be considered to alter the perception and reduce the visibility of the noise source and potentially allow for shielding. In addition to the reduction of disturbing sound, the presence of sounds with a positive connotation such as rustling leaves, water sounds or birdsong should be increased. Although it is difficult to completely mask negative sounds with positive ones (Van Renterghem, 2019), natural sounds such as birdsong or wind noise intensify the experience of nature and increase the restorative experience (Cameron et al., 2020). Future studies should specifically assess dose-response relationships between noise levels and restoration as well as the combined effects of road traffic noise and natural sounds. The findings further underscore the importance of noise-free periods (free from road traffic noise) in the macro-temporal pattern beyond the averaged noise metrics. Noise abatement strategies that typically focus on reducing overall sound pressure levels should also aim to increase both the frequency and duration of quiet periods.

The results regarding restoration, perceived restorativeness, and

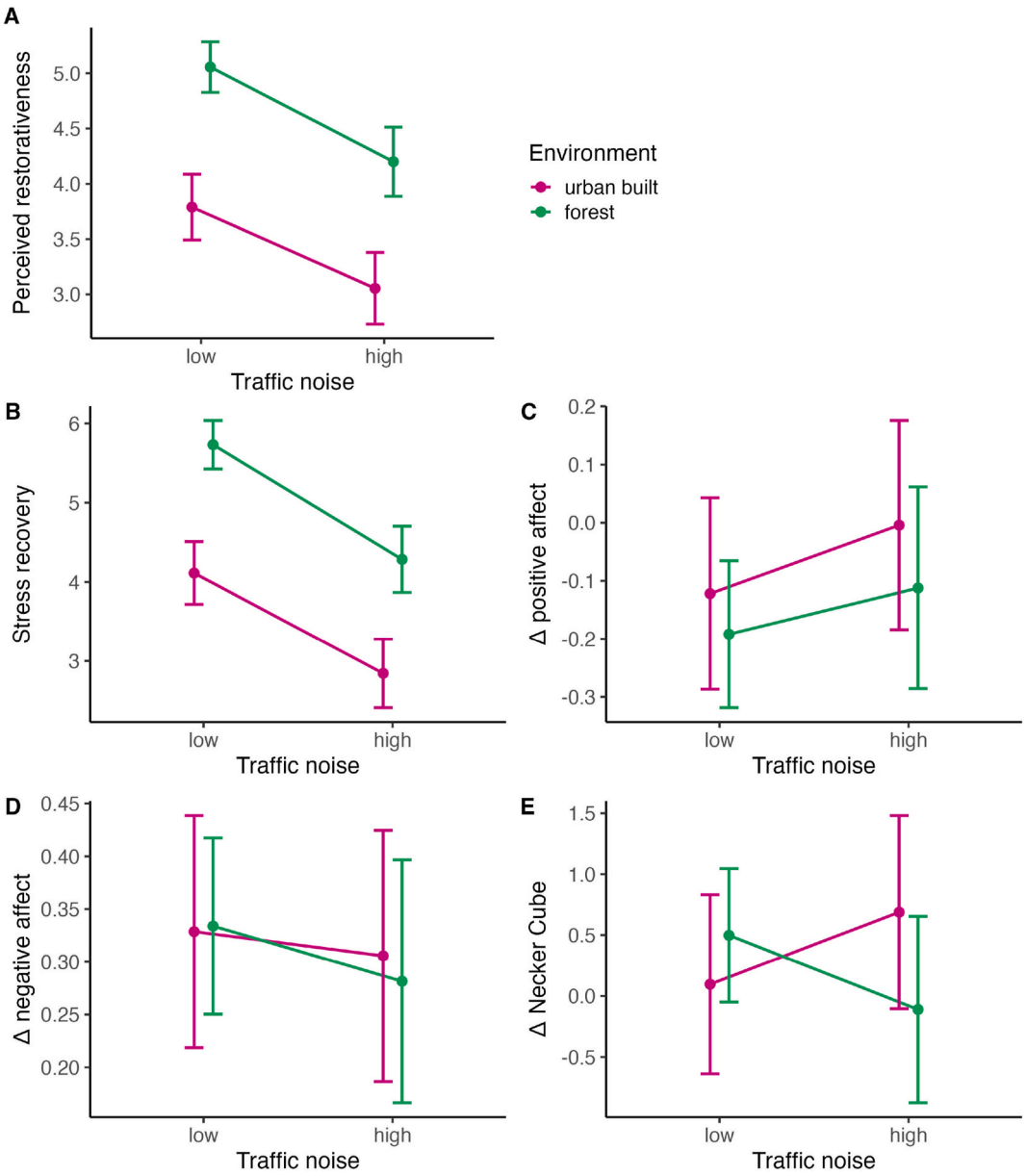


Fig. 5. Estimates for secondary outcomes.

subjective stress recovery additionally suggest that differences in the effects of exposure to greenspaces found in earlier studies (assessing these outcomes) may be confounded by disparate traffic noise exposure in the comparison settings. This might be especially true for studies comparing attractive greenspaces with unattractive urban built settings. This oversight may lead to misattributing effects to elements promoting restoration in natural environments that may rather be attributed to additional stressors in the comparison environments (see Bornioli & Subiza-Pérez, 2022; Scopelliti et al., 2019; Weber & Trojan, 2018 for more details). Our results thus highlight that the explicit assessment and inclusion of environmental stressors such as road traffic noise in the environments that are being compared in field experiments is necessary to clarify which environmental characteristics contribute to restoration and which hinder it. Continuous objective noise measurements during participant exposure should be employed, rather than relying on brief, isolated samples as seen in some previous studies (e.g. Gidlow et al., 2016). In addition, the analysis of the noise recordings from the walks in this study revealed that averaging noise metrics across entire recordings is not a sufficiently reliable proxy for road traffic noise, as recordings capture a mix of various sound sources. Future research should adopt our approach of extracting traffic noise more precisely by identifying time segments in which road traffic is the dominant sound source, rather than relying on overall averages that ignore source dominance—a limitation seen in prior studies (e.g. Gidlow et al., 2016). Moreover, we found that entirely excluding periods without dominant road traffic noise led to a systematic overestimation of the L_{Aeq} for walks with relatively little road traffic noise. As a methodological contribution, we therefore recommend including these non-dominant periods using a fixed substitute value—in our case, 30 dBA. This recommendation aligns with findings by Dopico et al. (2023), who observed discrepancies between measured and simulated road traffic noise in quieter areas. Finally, our study shows that a detailed analysis of the influence of traffic noise should include additional indicators, such as relative quiet time used here, beyond mere averaged noise metrics used in previous studies (Uebel et al., 2021) to refine the noise characterization.

Our findings further underscore the importance of thoroughly describing the compared environments and the challenge of selecting an appropriate comparison setting, as no neutral environmental baseline

Table 6
Effects for perceived restorativeness.

	Estimate	SE[b]	95 % CI[c]	p
Effect of environment averaged across noise conditions[a]	1.207	0.147	0.915, 1.498	<0.001
Effect of noise averaged across environment conditions	−0.795	0.145	−1.084, −0.506	<0.001
Effect of noise in urban built conditions	−0.735	0.216	−1.153, −0.317	0.001
Effect of noise in forest conditions	−0.856	0.194	−1.230, −0.482	<0.001
Age	−0.001	0.003	−0.007, 0.005	0.754
Gender (male)	−0.058	0.123	−0.301, 0.179	0.634
Interaction environment: noise	−0.121	0.290	−0.682, 0.441	0.678

Reference category for environment = urban, reference category for noise = low traffic noise. Reference category for gender = female.
[a] Calculated using custom contrasts.
[b] Standard error.
[c] CI = Confidence Interval.

Table 7
Effects for subjective stress recovery.

	Estimate	SE[b]	95 % CI[c]	p
Effect of environment averaged across noise conditions[a]	1.531	0.196	1.141, 1.920	<0.001
Effect of noise averaged across environment conditions	−1.358	0.194	−1.745, −0.972	<0.001
Effect of noise in urban built conditions	−1.270	0.289	−1.829, −0.711	<0.001
Effect of noise in forest conditions	−1.447	0.259	−1.946, −0.947	<0.001
Age	−0.013	0.004	−0.021, −0.005	0.002
Gender (male)	0.067	0.164	−0.252, 0.388	0.682
Interaction environment: noise	−0.177	0.388	−0.926, 0.573	0.650

Reference category for environment = urban, reference category for noise = low traffic noise. Reference category for gender = female.
[a] Calculated using custom contrasts.
[b] Standard error.
[c] CI = Confidence Interval.

Table 8
Effects on change in positive affect.

	Estimate	SE[b]	95 % CI[c]	p
Effect of environment averaged across noise conditions[a]	−0.089	0.081	−0.250, 0.072	0.447
Effect of noise averaged across environment conditions	0.099	0.080	−0.061, 0.259	0.447
Effect of noise in urban built conditions	0.118	0.120	−0.113, 0.348	0.328
Effect of noise in forest conditions	0.080	0.107	−0.127, 0.287	0.459
Age	0.002	0.002	−0.002, 0.006	0.251
Gender (male)	−0.017	0.074	−0.161, 0.127	0.816
Interaction environment: noise	−0.038	0.161	−0.348, 0.272	0.813

Reference category for environment = urban, reference category for noise = low traffic noise. Reference category for gender = female.
[a] Calculated using custom contrasts.
[b] Standard error.
[c] CI = Confidence Interval.

Table 9
Effects on change in negative affect.

	Estimate	SE[b]	95 % CI[c]	p
Effect of environment averaged across noise conditions[a]	−0.009	0.054	−0.116, 0.097	0.966
Effect of noise averaged across environment conditions	−0.038	0.054	−0.143, 0.068	0.966
Effect of noise in urban built conditions	−0.023	0.080	−0.180, 0.134	0.774
Effect of noise in forest conditions	−0.052	0.072	−0.193, 0.089	0.465
Age	−0.005	0.001	−0.007, −0.002	0.001
Gender (male)	−0.067	0.056	−0.177, 0.043	0.230
Interaction environment: noise	−0.029	0.107	−0.241, 0.182	0.784

Reference category for environment = urban, reference category for noise = low traffic noise. Reference category for gender = female.
[a] Calculated using custom contrasts.
[b] Standard error.
[c] CI = Confidence Interval.

exists against which exposure effects can be measured. Furthermore, they highlight the potential for future research on the restorative effects of pleasant urban built environments, like e.g. on effects of architectural elements such as building height.

We did not observe any interaction effects between environment and road traffic noise on restoration, perceived restorativeness or subjective stress recovery. This is a noteworthy finding in light of previous research on the congruence between environmental visuals and sound (e.g. Bruce & Davies, 2014). Individuals expect alignment between acoustic and visual characteristics, and incongruence—is typically perceived negatively (e.g. Carles et al., 1999). This could suggest that noise might be perceived more unfavorably in green spaces compared to urban built

Table 10
Effects on change in Necker Cube Score.

	Estimate	SE[b]	95 % CI[c]	p
Effect of environment averaged across noise conditions[a]	−0.199	0.360	−0.907, 0.509	>0.999
Effect of noise averaged across environment conditions	−0.008	0.356	−0.709, 0.693	>0.999
Effect of noise in urban built conditions	0.592	0.532	−0.454, 1.638	0.266
Effect of noise in forest conditions	−0.609	0.474	−1.541, 0.324	0.200
Age	−0.024	0.009	−0.042, −0.005	0.011
Gender (male)	0.599	0.371	−0.132, 1.329	0.108
Interaction environment: noise	−1.201	0.712	−2.601, 0.200	0.093

Reference category for environment = urban, reference category for noise = low traffic noise. Reference category for gender = female.
[a] Calculated using custom contrasts.
[b] Standard error.
[c] CI = Confidence Interval.

environments. However, our findings did not support this assumption.

Regarding salivary cortisol, our results align with previous studies applying a similar design (Gidlow et al., 2016; Rozario et al., 2024; Sudimac et al., 2024; Tyrväinen et al., 2014; Veitch et al., 2022) that found no significant difference in the change of salivary cortisol between urban and green conditions. Both our and previous studies observed a decrease in cortisol in both conditions, with no significant differences between conditions (Gidlow et al., 2016; Rozario et al., 2024; Sudimac et al., 2024; Tyrväinen et al., 2014; Veitch et al., 2022). Additionally, our findings are consistent with a meta-analysis indicating no significant effect of walking in urban versus natural environments on salivary cortisol levels (Mygind et al., 2021). In contrast, two studies included in the meta-analysis, Park et al. (2007) and Tsunetsugu et al. (2007), reported differences in salivary cortisol between conditions. However, both studies were rated as only moderately high quality in the meta-analysis (Mygind et al., 2021), relied on a very small sample size and exclusively included young male students aged 21–24 years, which may limit comparability to our sample. Additionally, baseline cortisol levels were unbalanced in both studies.

The challenge with salivary cortisol as a stress marker is distinguishing intervention effects from diurnal variations (Hunter et al., 2019). Miller et al. (2016) reported a typical diurnal decline in cortisol, which our study's decline (32.3 %–38.2 %) exceeds, suggesting that the decrease in our study reflects stress reduction from walking. However, we were not able to show effects due to environmental differences. This also aligns with Hunter et al. (2019), who observed a diurnal drop in a sample population of 11 % per hour between morning and nightfall. The road traffic noise exposure in this study of up to an L_{AE} of 98.8 dB on average (corresponding to an equivalent A-weighted continuous sound pressure level, L_{Aeq}, of 66.1 dB) did not change salivary cortisol levels significantly, consistent with other recent research findings (Kawai et al., 2024).

The inconsistent results regarding attention contradicted our hypotheses but align with literature. A meta-analysis by Stevenson et al. (2018) showed no effect of natural environments on attention control when considering studies with reported baseline balance. These

Table 11
Overview of results of three models with different noise predictors.

	Noise (low vs. high)			L_{AE}[c]			RQT[d]		
	Environment (urban vs. forest)[a]	Noise (low vs. high)[b]	Interaction environment: noise	Environment (urban vs. forest)	L_{AE}	Interaction environment: noise	Environment (urban vs. forest)	RQT	Interaction environment: noise
Restoration Outcome Scale	+*	-*	–	+*	-*	–	+*	+*	–
Cortisol	+	+	–	+	–	–	+	–	+
Perceived restorativeness	+*	-*	–	+*	-*	–	+*	+*	–
Subjective stress recovery	+*	-*	–	+*	-*	+	+*	+*	–
Positive affect	–	+	–	–	+	+	–	–	+
Negative affect	–	–	–	–	–	–	–	+	+
Attention	–	–	–	+	+	-*	+	–	+*

[a] Urban built as reference category.
[b] Low traffic noise as reference category.
[c] Sound exposure level.
[d] Relative quiet time.

Table 12
Acoustic characteristics in different conditions.

UBHT	FHT	UBLT	FLT
number of answers, %	number of answers, %	number of answers, %	number of answers, %
annoyed[a] (n = 9, 8.91 %)	disturbed (n = 13, 11.30 %)	calm (n = 11, 9.24 %)	calm (n = 48, 20.51 %)
angry (n = 7, 6.93 %)	stressed (n = 9, 7.83 %)	pleasant (n = 8, 6.72 %)	relaxed (n = 39, 16.67 %)
calm (n = 6, 5.94 %)	annoyed (n = 7, 6.09 %)	stressed (n = 7, 5.88 %)	joyful (n = 13, 5.56 %)
irritated (n = 6, 5.94 %)	calm (n = 7, 6.09 %)	relaxed (n = 6, 5.04 %)	peaceful (n = 13, 5.56 %)
stressed (n = 5, 4.95 %)	restless (n = 6, 5.22 %)	curious (n = 6, 5.04 %)	satisfied (n = 10, 4.27 %)

[a] Annoyed is included in the PANAS. All the other affects are not part of the PANAS. UBHT = urban built with high traffic noise, UBLT = urban built with low traffic noise, FHT = forest with high traffic noise, FLT = forest with low traffic noise.

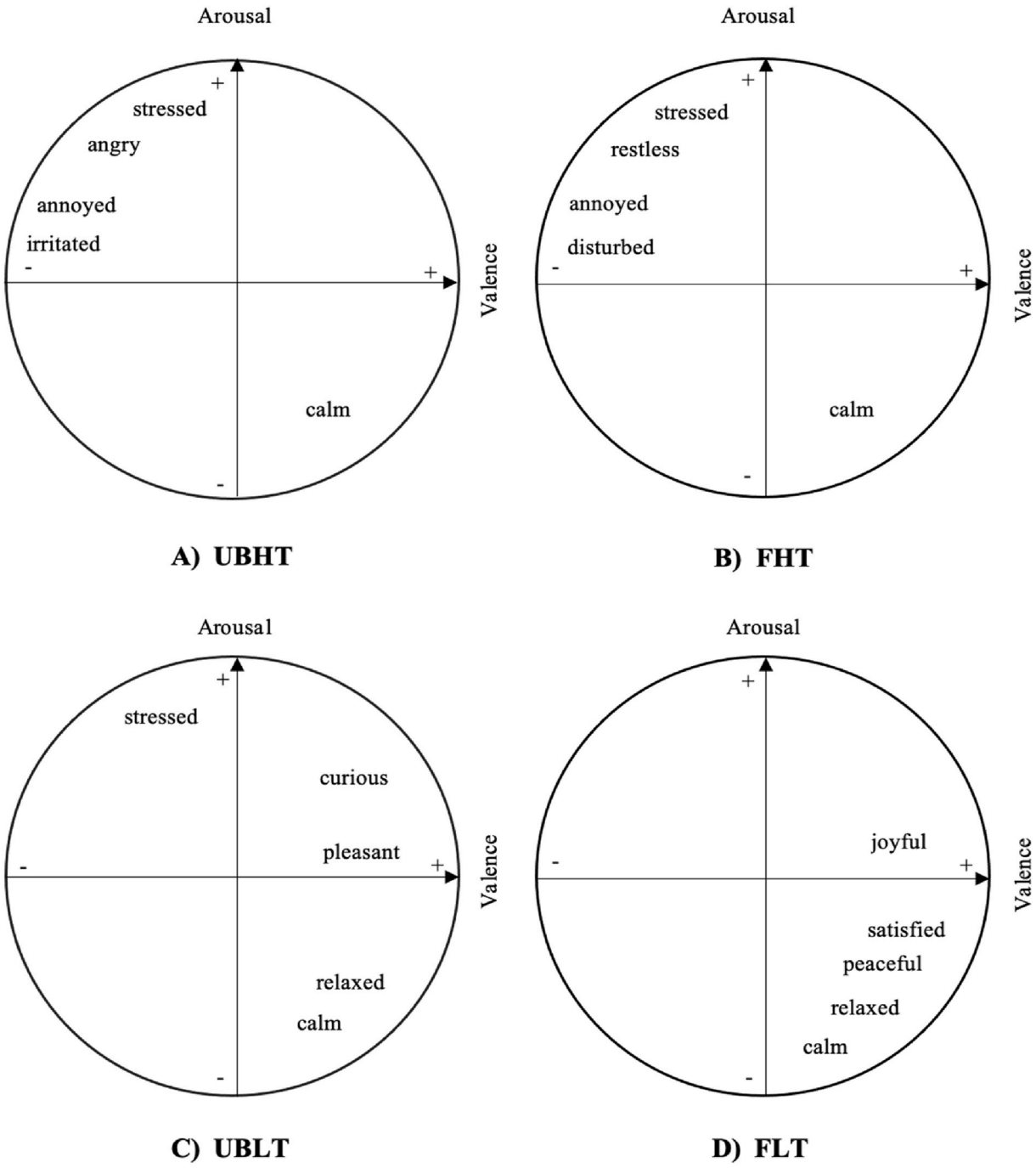


Fig. 6. Most frequently mentioned affects across conditions (open questions), arranged by the Circumplex Model of Affect. UBHT and FHT: Mostly high arousal negative affect items. UBLT: Affects scatter the most across valence and arousal dimensions. FLT: Low arousal positive valence items dominate.

included studies using the NCPCT (Hartig et al., 2003) and other measures of attention control (Attention Network Task, Stroop). Similarly, Ohly et al. (2016) found no difference in NCPCT scores after exposure to natural versus non-natural environments, drawing on the results of three studies (Cimprich & Ronis, 2003; Hartig et al., 2003; Tennessen & Cimprich, 1995). We also did not observe any clear effect of traffic noise on NCPCT scores. This is consistent with results from laboratory research (Kawai et al., 2024). The inconsistencies in our results, depending on the statistical model, likely arise from methodological weaknesses in the NCPCT. Future studies should consider more reliable measures, such as the Digit Span Backwards (DSB). However, a previous study by Stobbe and colleagues (2022) did not find effects of road traffic noise on cognitive performance in a laboratory experiment with an exposure duration of 6 min applying the Digit Span task.

Contrary to our hypotheses, significant differences in the PANAS were found neither between forest and urban built environments nor between environments with high and low traffic noise exposure. This contrasts with a meta-analysis by Yao et al. (2021) reporting that most studies reported an increase in positive affect and decreased negative affect after exposure to natural environments. The meta-analysis considered 20 studies, of which 18 used the PANAS. However, results showed high heterogeneity and low evidence quality. A meta-analysis by McMahan and Estes (2015) also found moderate increases in positive and decrease in negative affect through exposure to natural environments for studies using the PANAS (Berman et al., 2008, 2012; Mayer et al., 2009; Nisbet & Zelenski, 2011), but the designs of those

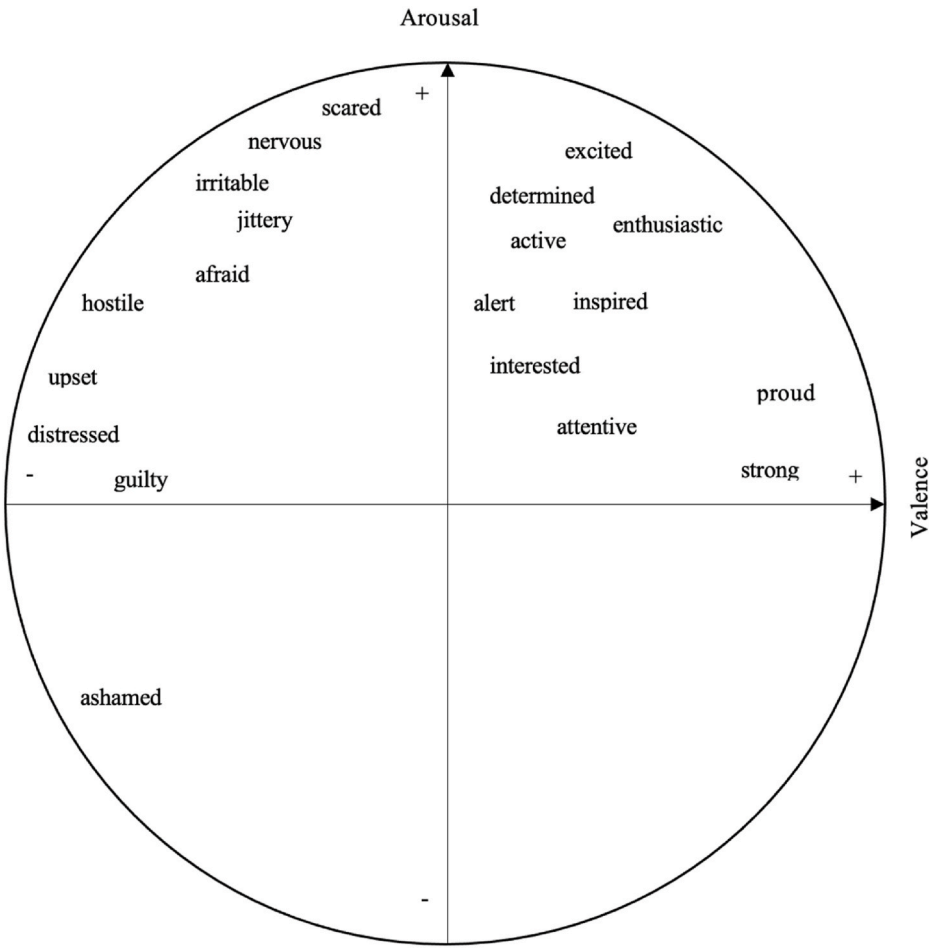


Fig. 7. PANAS items according to the Circumplex Model of Affect. High activation affect items dominate. Especially low activation positive valence items are missing.

studies—examining longer walks (Berman et al., 2008, 2012), only assessing affect post-exposure (Mayer et al., 2009; Nisbet & Zelenski, 2011), or using a modified version of the PANAS (Nisbet & Zelenski, 2011)—differed from ours. More recent studies using the PANAS show mixed results (Rozario et al., 2024; Sudimac et al., 2024; Tyrväinen et al., 2014). Interestingly, the findings for the open question in this study show a more distinct picture that better aligns with the literature. Overall, the high arousal negative affect items dominated in the conditions with high road traffic noise, while low arousal positive valence items dominated in the forest with low road traffic noise. In the urban condition with low traffic noise, mentioned affects were the most scattered across the dimensions; it was also the condition with the most diverse soundscape. In sum, valence and arousal discriminated between affective reactions to sound environments with differing extents of road traffic noise, which is consistent with previous studies (Västfjäll et al., 2003). In the PANAS, items with high arousal dominate, and low activation items (especially those with positive valence) are missing (see Isbell & Lair, 2020). Since the latter were shown by the open question to be decisive for describing differences in the affective experience of walking in disparate environments, this reveals a potentially important limitation of the PANAS and suggests a need to develop and use affective measures that include low activation affect (Harmon-Jones et al., 2016). One affect measure for which scale ratings were shown to be able to be represented as a circumplex structure with the dimensions of valence and activation is the Swedish Score Affect Scale (Västfjäll et al., 2002).

Regarding covariate effects, older participants experienced lower restoration and subjective stress recovery, smaller decreases in cortisol and negative affect, and a smaller increase in attention. This is likely because they arrived at the test sites with lower initial stress levels, as confirmed by our data showing lower baseline cortisol. One possible explanation is that participants who are retired may experience less stress and time pressure compared to younger individuals. As a result of the lower initial stress levels, older participants had less potential for restoration. This aligns with literature describing instoration as a resource-building process similar to restoration, but weaker that does not require exposing participants to a stressor at the outset of an experiment (Hartig, 2007; Korpela & Ratcliffe, 2021).

4.2. *Strengths and limitations*

This is one of the first field studies to consider and systematically include road traffic noise exposure in different environments as a factor potentially affecting instoration. The factorial design allowed us to separate the contributions of setting (urban vs. forest) and noise exposure (high vs. low) to the outcomes, which is an additional asset. A particularly noteworthy aspect is the detailed extraction of road traffic noise from the audio recordings, by analyzing very short (1-s) time segments to determine whether traffic noise was the dominant sound. Another strength is the inclusion of multiple settings per condition, reducing the risk of systematically conflating loud and quiet as well as forest and urban built settings with other setting variables.

Nevertheless, some limitations must be considered. First, although we controlled for road traffic noise, it was still highly variable within test sites, and the same sound exposure for “quiet” and “loud” could not be obtained for the urban and forest environments. However, since noise was also assessed continuously (L_{AE}, RQT), and the results of all three

models were largely consistent, it is unlikely that the systematic differences affected the results. Second, varying nature sounds during walks could not be controlled, and future laboratory studies should investigate the combined effects of road traffic noise and natural sounds. Third, other potentially confounding factors like air pollution and visual landscape complexity may differ between settings and were not captured. Fourth, covariates such as residential exposure to traffic noise and green, which might influence restoration, were not assessed and should be captured in future studies. Fifth, while mixed-effects models are generally robust to unbalanced designs, the higher number of participants in the forest with low traffic noise condition may have affected the precision of estimates, leading to more stable estimates in that condition. Sixth, no inactive control (i.e., sitting on a bench) was included in our study. The positive effects of walking itself may have masked potentially less dominant effects of road traffic noise and/or vegetation on certain outcomes. Future studies should include an inactive control to isolate the effects of walking from traffic noise and vegetation. Seventh, we cannot rule out the possibility of a self-selection bias regarding which participants took part in our study. Individuals participating in a walking study may already have preconceived beliefs about the likely effects of walking, as well as preferences for specific environments, even if they are unaware of the environments included in the other conditions.

5. Conclusions

The randomized, controlled trial comparing the effects of walking in forests and urban built environments with high or low road traffic noise exposure was aimed at providing a better empirical basis for: (1) the practical application of walking and nature-based interventions as low-threshold approaches to promoting recovery from stress and improving well-being; (2) the impact of additional environmental characteristics of greenspaces and urban built environments, such as road traffic noise, as an impediment to stress recovery; and (3) urban planning and the creation and maintenance of public greenspaces and restorative environments that support the well-being of residents. It revealed that, overall, walking in forests, in environments with lower road traffic noise or with higher RQT, was associated with increased feelings of restoration, subjective stress recovery, and perceived restorativeness, compared to walking in urban environments and such with higher road traffic noise or less RQT. These associations, however, were not observed for the stress biomarker salivary cortisol or for attention, though cortisol levels decreased in all conditions, likely due to walking itself. Findings from quantitative and qualitative data on soundscape-induced positive and negative affect deviated, hinting at shortcomings of the PANAS for assessing these differences and indicating a need for developing and using affective measures that include low activation affect.

Specifically, our study highlighted the importance of elevated road traffic noise of an L_{Aeq} up to some 66 dB to significantly impair subjective feelings of restoration, stress recovery, and perceived restorativeness. Thus, when planning, constructing, and preserving environments that support well-being, the negative effects of road traffic noise on subjective restoration and subjective stress recovery should be considered. This highlights the need for noise mapping and mitigation strategies to reduce noise exposure, particularly in urban forests and public greenspaces, to fully harness their potential to promote restoration. It further emphasizes the importance of incorporating noise protection for recreational areas into law. The study further highlights the importance of noise-free periods (free from road traffic noise) in the macro-temporal pattern beyond the averaged noise metrics. Future research comparing the benefits of exposure to greenspaces and urban built environments should explicitly consider traffic noise as one of many other potential environmental stressors.

CRediT authorship contribution statement

Julia Schaupp: Writing – review & editing, Writing – original draft, Visualization, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Beat Schäffer:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization. **Jean-Marc Wunderli:** Methodology, Conceptualization. **Karin Hediger:** Writing – review & editing, Supervision. **Silvia Tobias:** Funding acquisition. **Jakob Heimer:** Validation. **Nicole Bauer:** Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Data availability

The data are publicly available on the Open Science Framework (https://osf.io/2bfgv/).

Declaration of generative AI and AI-assisted technologies in the writing process

During preparation of this work, JS used ChatGPT to shorten the manuscript. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Funding sources

This research is funded by the Swiss National Science Foundation (SNF Research Grant Number CRSII5_193847). Additionally, xy is funded through an Eccellenza grant (PCEFP1_194591) from the Swiss National Science Foundation.

Declaration of competing interest

The authors declare no competing interests.

Acknowledgments

The authors are grateful to the participants of the study, to the civil servants and research assistants helping with data collection, and to Markus Haselbach for supporting with the noise measurements. Further, we thank Terry Hartig for his extremely helpful comments and shared ideas on the contrasting results of the PANAS and the qualitative data on soundscape-induced positive and negative affect.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jenvp.2025.102662.

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